

Adherence to AAP Guidelines for Well-Child Care Under Managed Care

Robert S. Byrd, MD, MPH*; Robert A. Hoekelman, MD†; and Peggy Auinger, MSS

ABSTRACT. *Objectives.* To determine adherence to American Academy of Pediatrics (AAP) Recommendations for Preventive Pediatric Health Care in Monroe County, New York by individual patients and individual pediatricians under managed care practice and to compare adherence-to-recommendations rates for privately insured and publicly funded managed care patients.

Study Designs and Methods. Using claims data for children 0 through 18 years of age cared for by pediatricians, we compared adherence to well-child care (WCC) visit recommendations for 130 572 children enrolled in a privately insured managed care system during 1992, 1993, and 1994 to 17 586 children insured by a publicly funded, Medicaid-managed care system during 1994 and 1995. Criteria for WCC visit adherence were based on 1991 AAP guidelines of 19 office visits from birth through 18 years of age. Adherence-to-recommendations rates by individual pediatricians also were determined.

Results. Despite complete financial coverage of WCC visits (with no co-payment or deductible charges) by both insurance systems, strict adherence to AAP guidelines for WCC visits was low. Only 46% of privately insured and 35% of publicly funded children received all the recommended visits during the study period. During the same period, 17% of privately insured and 35% of publicly funded managed care patients received no WCC. There was little difference in the rate of full WCC visit adherence by age in either system with the rates ranging in privately insured patients from 49% in infants (<2 years of age) to 47% in adolescents (12 through 18 years of age) and ranging in publicly funded patients from 36% to 34% in these two age groups, respectively. Only 2% of privately insured infants had no record of WCC compared with 29% of adolescents. This contrasted with 12% of infants and 54% of adolescents who were publicly funded. Of pediatricians, <5% achieved 100% adherence to AAP guidelines for their patients (privately insured or publicly funded). Pediatricians completed an average of 52% of the recommended visits with their publicly funded patients and 68% of the recommended visits with their privately insured patients.

Conclusions. WCC visits were underutilized for children in both managed care systems. Children of parents who have low incomes presumably could benefit greatest by preventive visits, but these children were less likely to receive the recommended number of WCC visits. Finding ways to increase the number of WCC visits

that all children make is a major challenge, as is conducting studies that prove their worth. *Pediatrics* 1999;104:536–540; *well-child care, managed care, Medicaid.*

ABBREVIATIONS. AAP, American Academy of Pediatrics; WCC, well-child care.

Since 1967, the American Academy of Pediatrics (AAP) has made recommendations for preventive pediatric health care in which guidelines for the frequency and content of well-child care (WCC) visits have been provided. These recommendations have been based on observed practices and expert opinion and have been revised seven times (1974, 1977, 1982, 1985, 1987, 1991, and 1995). In recent years, the recommendations have been formulated by the AAP Committee on Practice and Ambulatory Medicine. The latest revision was generated initially by a special panel of 65 distinguished child health experts representing 16 national professional organizations and sponsored by the Maternal and Child Health Bureau of the US Public Health Service and the Medicaid Bureau of the Health Service Administration. The panel published its recommendations, entitled *Bright Futures*,¹ in 1994; they were adopted by the AAP in 1995.² Those recommendations include the number of WCC visits to be made from birth through 21 years of age and the components of care that should be conducted at each visit. The number of WCC visits (after discharge from the newborn nursery) that have been recommended has grown from 14 to 27 between 1967 and 1995. The number of WCC components recommended for each visit over this 28-year span also has increased substantially.

The purpose of this study was to determine the adherence to the AAP recommendations for WCC visits by individual patients and pediatricians in our community, Monroe County, New York, under managed care practice and to compare adherence-to-recommendations rates for privately insured and publicly funded children.

METHODS

The study was conducted using office-based health care claims data for children 0 through 18 years of age living in Monroe County, New York, with the city of Rochester at its center. The children were either enrolled in a private health maintenance organization or in a public Medicaid-funded program, both of which provided managed health care. Adherence to AAP guidelines for office-based WCC from birth through 18 years of age was the major outcome of interest; thus, analyses were limited to children who received primary care by pediatricians.

In the private health maintenance organization system in this

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study, 130 572 children received primary care by pediatricians during 1992, 1993, and 1994. They represent >70% of the total child population in this age group for the county (1990 census data estimated the population 0 through 18 years of age to be 185 778). The claims data for the public system were drawn from 1994 and 1995 claims records and involved all 17 586 children insured under the county's managed care, Medicaid system. During these years, both private and public systems were fee-for-service. Table 1 shows that the fee schedules were slightly higher for public care for infancy through adolescence but were significantly higher for private care provided to patients over 18 years of age. Neither involved co-payment or deductible charges. Details regarding the data collected by both systems were comparable, because both used similar claims forms, and the data were entered into a similar computer record. However, the data did not overlap completely in terms of the years reported; the private system did not have 1995 data available at the time of our analysis, and the public system did not have comparable data until 1994.

Criteria for WCC visit adherence were based on 1991 AAP guidelines for 19 designated office visits through 18 years of age, rather than on the 1995 guidelines, because the latter were not in effect during the years that the visits were made. The cut-off at 18 years of age was used at the request of both managed care systems, because 19-, 20-, and 21-year-old patients were tracked differently from younger patients in their claims data system. The adherence to these guidelines was calculated for each child based on the first and last visits recorded in the claims data. A WCC visit during the age ranges presented in Table 2 would be counted as 100% adherence with the AAP recommendations for the particular designated visit. Adherence for each child was determined by comparing the number of WCC visits made at the designated intervals with the number that should have been made during the time that spanned the child's first and last visits. A WCC visit was determined solely by ICD-9 codes for patient encounters. Comparisons with immunization procedure codes provided an internal check of the validity of ICD-9 codes to determine when WCC visits occurred. Otitis media was the most frequently listed diagnosis for which an immunization was given without a designation of WCC, but only accounted for 2.5% of visits for the publicly insured children and 1.6% of visits for privately insured children. Otitis media accounted for more than half of such visits. These visits did not account for a significant proportion of visits in which one component (immunization) of WCC was delivered. Furthermore, in visits in which an acute condition is the primary reason for the visit, we doubt all the components of a well-child visit are covered systematically, even if vaccinations are delivered. Thus, non-WCC visits in which immunizations and anticipatory guidance might have been delivered in the course of treating an acute condition, such as a viral upper respiratory infection, otitis media, or asthma, were not considered as WCC visits in this study.

Encounter data were used in this study. Some patients had only one visit during the study, and although they might have been registered for service the entire study period, they would be considered to have 100% compliance if that one visit was for WCC, and 0% if it was not. There were probably some insured children who did not have any encounters during the study period; they did not show up in our data because registration data were not available for study. Therefore, this approach overestimates the rate of adherence to WCC guidelines because registered children with no visits are not counted. Furthermore, we chose not to confine the study to those children who were enrolled continuously because we did not have registration data to define clearly the period of enrollment. In addition, excluding patients who were not enrolled continuously would probably select outpatients who

TABLE 2. WCC Visits Recommended by the AAP, 1991: Age Range of Children For Each Recommended Visit

AAP Recommended Visit	Age Range
By 1 mo	7 d to <1 mo
2 mo	1–2 mo
4 mo	3–4 mo
6 mo	5–7 mo
9 mo	8–10 mo
12 mo	11–13 mo
15 mo	14–16 mo
18 mo	17–19 mo
24 mo	20–29 mo
3 y	30–41 mo
4 y	42–53 mo
5 y	54–65 mo
6 y	66–83 mo
8 y	84–107 mo
10 y	108–131 mo
12 y	132–155 mo
14 y	156–179 mo
16 y	180–203 mo
18 y	204–227 mo

are more likely to receive fewer than the recommended number of WCC visits.

The encounter data included a designation of the primary care provider at the time of an outpatient visit. Although some patients may have had multiple providers during the study period, the rate of adherence to WCC guidelines for any given provider was calculated by taking the number of WCC visits that children had during the period of their care and dividing that by the number of WCC visits that were indicated by the guidelines during that same period.

Public-private system comparisons of the adherence to WCC guidelines were made. In addition to determining the adherence to WCC guidelines for individual children, adherence to AAP recommendations were determined for individual pediatricians by comparing the number of WCC visits made by children for whom they were the designated primary care practitioner to the number of WCC visits that should have occurred based on AAP recommendations. χ^2 tests were used to test for differences in proportions.³

Because the adherence rates for privately insured patients spanned only 3 years and those for publicly funded patients only 2 years, they are cross-sectional rather than longitudinal. Therefore, adherence rates for patients and pediatricians for the 19 recommended visits spanning 18 years could not be determined in this study. Nevertheless, because children of all ages were included in the data analyzed, there is no reason to suppose that the cross-sectional adherence rates determined would be different from longitudinal adherence rates for this population.

RESULTS

Despite complete financial coverage of WCC visits by both insurance systems, strict adherence to AAP guidelines for WCC visits was low. As seen in Fig 1, only 46% of privately insured and 35% of publicly funded managed care children received all the recommended visits during the study period. During the same period, 17% of privately insured and 35% of publicly funded patients received no WCC.

Figure 2 shows that adolescents were more likely than were younger children to have made no WCC visits in both managed care systems. Children in publicly funded managed care across all ages were more likely to have received no WCC. Only 2% of privately insured infants had no record of WCC, compared with 29% of adolescents ($P < .01$), contrasted with 12% of infants and 54% of adolescents who were publicly funded ($P < .01$). Figure 3 displays age-specific rates of children with 100% WCC

TABLE 1. 1995 Fee Schedule for WCC Visits Under Public and Private Managed Care Systems in Monroe County, New York

Age Group	Age (Years)	Managed Care System	
		Public (\$)	Private (\$)
Infancy	0–1	44.19	38.24
Early childhood	1–4	46.65	42.15
Late childhood	5–11	46.65	42.49
Adolescence	12–17	51.56	46.94
Adults	18+	63.83	90.75

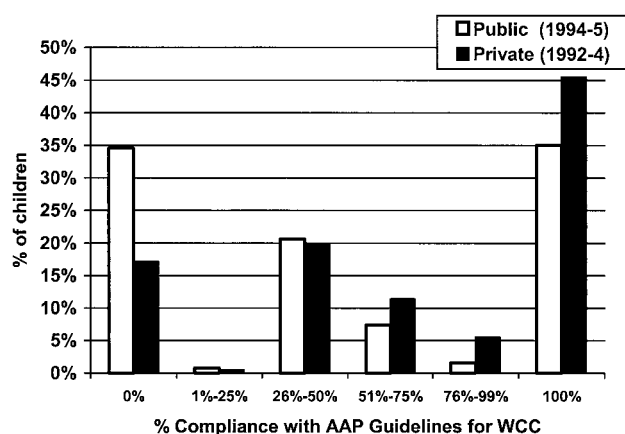


Fig 1. Comparison of adherence to AAP-WCC visit guidelines under public and private managed care systems for children 0 through 18 years of age in Monroe County, New York.

compliance during the study period. There was little difference in the rate of full WCC visit adherence by age in either system, especially when considering only those children who received any WCC, as is shown in Fig 3. Among all children, the rates of full compliance range in privately insured patients from 49% in infants (<2 years of age) to 47% in adolescents (12-18 years of age) and range in publicly funded patients from 36% to 34% ($P = .26$) in these two age groups, respectively (not shown in figures). When considering only those children and adolescents who received some WCC, differences between sources of insurance funding remained small, and rates of full compliance were higher among adolescents, for whom only 1 well-child visit may have put them into full compliance (Fig 3).

Pediatricians also were compared in both systems, as shown in Fig 4. Of the pediatricians, <5% achieved 100% adherence to AAP, WCC visit guidelines for all their patients. Likewise, <5% of pediatricians listed as a child's primary care physician (privately insured or publicly funded) never saw the child for WCC. Most were somewhere in the middle. What is striking is that two bell-shaped curves emerge from these data. Pediatricians completed an average of 52% of the recommended visits with their publicly funded patients and 68% of the recom-

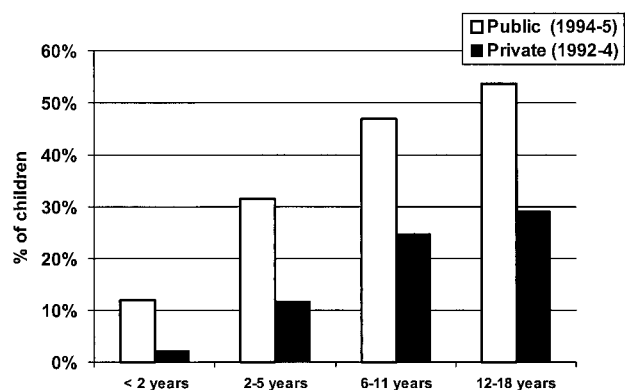


Fig 2. Percentage of children 0 through 18 years of age who had no WCC visits under public and private managed care systems in Monroe County, New York.

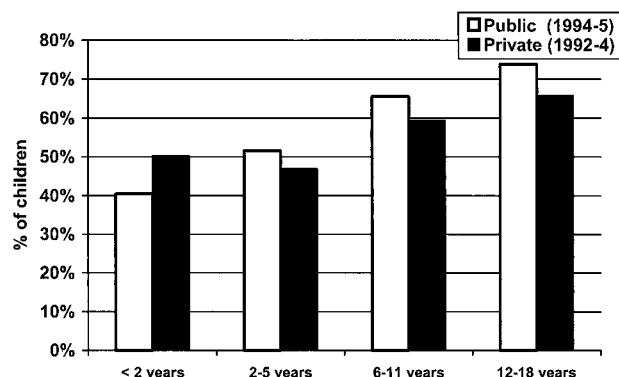


Fig 3. Percentage of children 0 through 18 years of age who received any WCC and for whom AAP guidelines were adhered to fully during WCC visits under public and private managed care systems in Monroe County, New York.

mended visits with their privately insured patients. Although some pediatricians saw mostly public or mostly private patients, many saw a sizable portion of patients covered by both types of managed care insurance. For those pediatricians who saw patients who had both types of insurance, discrepancies in adherence between the two groups existed with privately insured patients consistently receiving a greater proportion of recommended WCC visits than did publicly funded patients.

DISCUSSION

The purpose of this study was to determine how well patients (parents) and primary care pediatric practitioners in our community adhered to the AAP Recommendations for Preventive Pediatric Health Care when the entire cost of WCC visits was underwritten by managed care insurance programs.

The objectives for WCC visits have been 1) the prevention of disease and the promotion of health through immunizations and health education; 2) the early detection and treatment of disease through history taking, physical examinations, and screening for specific diseases; and 3) the provision of anticipatory guidance in all aspects of child rearing. Attaining these objectives is designed to lower mortality, reduce morbidity and disability, promote optimum growth and development, and help children achieve longer, fuller, and more productive lives. These are compelling reasons for both parents and pediatricians to adhere to the AAP recommendations.

Parental Adherence

Parents most likely are not fully cognizant of the objectives for WCC visits but could be motivated to make visits for other reasons. Included among these reasons are 1) gaining reassurance about their child's health status; 2) obtaining answers to specific questions about child rearing; 3) receiving approval of their parenting skills; 4) doing what is expected of them (if they have been told by their pediatrician); 5) meeting socialization needs (by providing social interactions with nurses, doctors, and other parents immediately and by talking about them afterward with friends, neighbors, and relatives); and 6) estab-

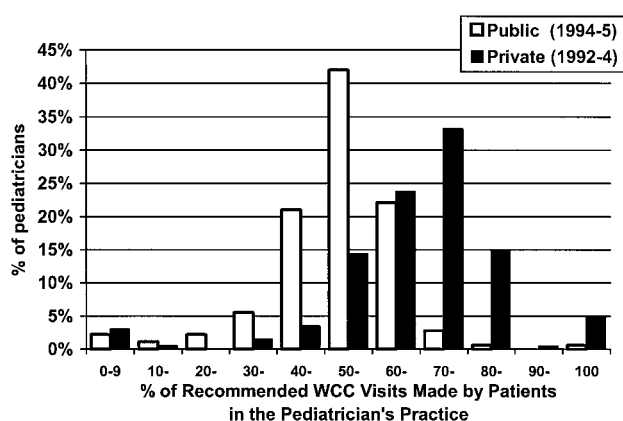


Fig 4. Pediatrician-specific adherence to AAP guidelines for WCC visits under public and private managed care systems in Monroe County, New York.

lishing a source of care when their child becomes ill. Despite these incentives, most parents do not adhere to the AAP-recommended WCC visits schedule even when the costs of WCC visits are insured fully, as demonstrated in this and other studies. This study did not examine parental attitudes or behaviors that would provide insights into the contribution of these factors in adherence. The major difference between patients who have publicly funded managed care health insurance and those with privately funded insurance in this study was the proportion of patients who received no WCC. When comparisons were limited to those who received some WCC, differences between adherence rates based on sources of funding were negligible.

Some studies have demonstrated that mothers who have the following characteristics do not comply with AAP-recommended WCC visit schedules: 1) they are employed; 2) they are raising their child alone; 3) they are poor; 4) they are young; 5) they change their residence frequently; 6) they did not use prenatal care adequately; 7) they are members of a minority group; and 8) they are poorly educated.⁴⁻⁶ In a study of white, middle-class families with well-educated parents enrolled in a university-affiliated prepaid group practice, Horwitz et al⁷ found that the highest utilization rates for WCC visits were for younger children and first-born children. Woodward et al⁴ confirmed these findings and also found that when a child's physical health status as perceived by the parents was poor, all ambulatory medical care use including preventive services increased. Our study did not examine these maternal or child characteristics.

Although absence of cost to the family of well-child visits does not ensure adherence to AAP-recommended WCC visit schedules, adherence is reduced when full or partial fees for WCC visits are charged.⁸⁻¹¹

Pediatrician Adherence

It is difficult to determine why most primary care pediatricians do not do more to ensure that their patients receive the AAP-recommended number of WCC visits. Mustard et al⁶ and Orr et al¹¹ relate the

use of health services including WCC visits to the organization and delivery of those services suggesting that if these elements were improved and were made more user friendly, adherence by parents would improve. The amount of effort that the AAP has expended in generating its recommendations for preventive pediatric health care should dictate to its members to do more than they have to adhere to them. However, in a 1992 survey of AAP members, only 59% of respondents reported that they followed the AAP-recommended WCC visit schedule.¹² Perhaps the experience of those who do not indicates to them that fewer WCC visits are needed than are recommended, or they understand that the number of recommended visits has not been proven to be of sufficient benefit given the time and effort they require. These views have been stated often in the literature.¹³⁻¹⁸ Despite such reservations on the part of primary care pediatric practitioners and others over the years, the number of AAP-recommended WCC visits has increased steadily. Proving the effectiveness of WCC recommendations by the AAP has been called for by many but has never been accomplished.^{19,20} Studies of their worth are long overdue.

This study demonstrates great variability in adherence rates among pediatricians within each insurance type. Such findings suggest that some pediatricians can do more to improve adherence to WCC guidelines, if one can determine how the best practices differ from those with lower adherence rates.

There is no reason to think that adherence to the AAP-recommended WCC visit schedule under private and public managed care insurance programs demonstrated in this study for Monroe County, New York varies significantly in other US communities. This study, however, was conducted in a fee-for-service care setting in which incentives to providers should have encouraged better compliance with AAP guidelines. However, one limitation of this study should be considered. The study was conducted using encounter data, and the number of well child visits that occurred within the time frame of the patients' first and last visit was used to determine adherence rates. A more precise measure of adherence to AAP-WCC guidelines might have been obtained by using registration data to determine the exact period of insurance coverage as the time frame to calculate adherence. If these data had been available for this study, it likely would have lowered the rates of compliance with WCC guidelines. Additionally, this study does not take into account the possibility that a child received WCC immediately before the start of the study period and thus may underestimate the true rate of adherence. Although these two factors may affect the precision of the estimate, there is no reason to believe that it would produce a systematic bias in the results presented in this study. To increase the precision of an estimate of adherence to WCC guidelines, one might limit the sample to those children who were enrolled continuously for a specified time frame. This approach, however, will eliminate from analyses children who go on and off insurance coverage. This at-risk group that would be excluded from analyses and consideration might be

one that warrants specific targeting for health supervision and anticipatory guidance.

Monitoring rates of compliance to WCC may serve as one benchmark for how well children, especially those living in poverty, will be served by a capitated care system. It is particularly disturbing that children of low-income families, who presumably could benefit more from preventive health care services than could children of higher socioeconomic status, receive fewer such services. Finding ways to increase the number of WCC visits that all children make is a major challenge for their primary care pediatricians and the health care systems in which they practice.

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